

Table 6.1: Main keywords that can be used in the simple input of ORCA.

Keyword	Input block	Variable	Comment
HF DFT FOD	METHOD	METHOD	Selects the Hartree–Fock method Selects the DFT method (see table 6.2 on page 23 for a list of functionals) FOD analysis (see 9.6.8.2) employing default settings (TPSS/def2-TZVP, TightSCF, SmearTemp = 5000 K)
Runtypes			
ENERGY or SP OPT COPT ZOPT GDIIS-COPT GDIIS-ZOPT GDIIS-OPT ENGRAD NUMGRAD NUMFREQ NUMNACME MD CIM	METHOD	RUNTYP	Selects a single point calculation Selects a geometry optimization calculation (using internal redundant coordinates) Optimization in Cartesian coordinates (if you are desperate) Optimization in Z-matrix coordinates (dangerous) COPT using GDIIS ZOPT using GDIIS Normal optimization using GDIIS Selects an energy and gradient calculation Numerical gradient (has explicitly to be asked for, if analytic gradient is not available) Numerical frequencies Numerical non-adiabatic couplings (only for CIS/TD-DFT) Molecular dynamic simulation Cluster-In-Molecule calculation
Atomic mass/weight handling			
Mass2016	METHOD	AMASS	Use the latest (2016) atomic masses of the most abundant or most stable isotopes instead of atomic weights.
Symmetry handling			
UseSym NoUseSym			Turns on the use of molecular symmetry (see section 6.5). <i>THIS IS VERY RUDIMENTARY!</i> Turns symmetry off
Second order many body perturbation theory			
MP2 MP2RI or RI-MP2 SCS-MP2 RI-SCS-MP2			Selects Method=HF and DoMP2=true Select the MP2-RI method Spin-component scaled MP2 Spin-component scaled RI-MP2 (synonym is SCS-RI-MP2)

OO-RI-MP2 OO-RI-SCS-MP2 MP2-F12 MP2-F12-RI MP2-F12D-RI			Orbital optimized RI-MP2 Orbital optimized and spin-component scaled RI-MP2 MP2 with F12 correction (synonym is F12-MP2) MP2-RI with RI-F12 correction MP2-RI with RI-F12 correction employing the D approximation (less expensive) (synonyms are F12-RI-MP2, RI-MP2-F12)
High-level single reference methods. These are implemented in the MDCI module. They can be run in a number of technical variants.			
CCSD CCSD(T) CCSD-F12 CCSD(T)-F12 CCSD-F12/RI CCSD-F12D/RI CCSD(T)-F12/RI CCSD(T)-F12D/RI QCISD QCISD(T) QCISD-F12 QCISD(T)-F12 QCISD-F12/RI QCISD(T)-F12/RI CPF/1 NCPF/1 CEPA/1 NCEPA/1 RI-CEPA/1-F12 MP3 SCS-MP3	MDCI	CITYPE	Coupled-cluster singles and doubles Same with perturbative triples correction CCSD with F12 correction CCSD(T) with F12 correction CCSD with RI-F12 correction CCSD with RI-F12 correction employing the D approximation (less expensive) CCSD(T) with RI-F12 correction CCSD(T) with RI-F12 correction employing the D approximation (less expensive) Quadratic Configuration interaction Same with perturbative triples correction QCISD with F12 correction QCISD(T) with F12 correction QCISD with RI-F12 correction QCISD(T) with RI-F12 correction Coupled-pair functional A “new” modified coupled-pair functional Coupled-electron-pair approximation The CEPA analogue of NCPF/1 RI-CEPA with F12 correction MP3 energies Grimme’s refined version of MP3
Other coupled-pair methods are available and are documented later in the manual in detail (section 7.8) In general you can augment the method with RI-METHOD in order to make the density fitting approximation operative; RI34-METHOD does the same but only for the 3- and 4-external integrals). MO-METHOD performs a full four index transformation and AO-METHOD computes the 3- and 4-external contributions on the fly. With AOX-METHOD this is done from stored AO integrals.			

Local correlation methods. These are local, pair natural orbital based correlation methods. They must be used together with auxiliary correlation fitting basis sets. Open-shell variants are available for some of the methods, for full list please see section 8.1.3. We recommend $n = 1$ for the CEPA methods.			
LPNO-CEPA/ n	MDCI	Various	Local pair natural orbital CEPA methods
LPNO-CPF/ n			Same for coupled-pair functionals
LPNO-NCEPA/ n			Same for modified versions
LPNO-NCPF/ n			Same for modified versions
LPNO-QCISD			Same for quadratic CI with singles and doubles
LPNO-CCSD			Same for coupled-cluster theory with single and double excitations
DLPNO-CCSD			Domain based local pair natural orbital coupled-cluster method with single and double excitations (closed-shell only)
DLPNO-CCSD(T)			DLPNO-CCSD with perturbative triple excitations
DLPNO-CCSD(T1)			DLPNO-CCSD with iterative perturbative triple excitations
DLPNO-MP2	MP2	Various	Local (DLPNO) MP2
DLPNO-SCS-MP2			Spin-component scaled DLPNO-MP2 (a synonym is SCS-DLPNO-MP2)
DLPNO-MP2-F12			DLPNO-MP2 with F12 correction employing an efficient form of the C approximation
DLPNO-MP2-F12/D			DLPNO-MP2-F12 with approach D (less expensive than the C approximation)
DLPNO-CCSD-F12			DLPNO-CCSD with F12 correction employing an efficient form of the C approximation
DLPNO-CCSD-F12/D			DLPNO-CCSD-F12 with approach D (less expensive than the C approximation)
DLPNO-CCSD(T)-F12			DLPNO-CCSD(T) with F12 correction employing an efficient form of the C approximation
DLPNO-CCSD(T)-F12/D			DLPNO-CCSD(T)-F12 with approach D (less expensive than the C approximation)
DLPNO-NEVPT2			DLPNO-NEVPT2 requires a CASSCF block

Accuracy control for local correlation methods. These keywords select predefined sensible sets of thresholds to control the accuracy of DLPNO calculations. See the corresponding sections on local correlation methods for more details.			
LoosePNO NormalPNO TightPNO DLPNO-HFC1 DLPNO-HFC2	MDCI, MP2	Various	Selects loose DLPNO thresholds Selects default DLPNO thresholds Selects tight DLPNO thresholds Tightened truncation setting for DLPNO-CCSD hyperfine coupling constants calculation Tighter truncation setting than for DLPNO-HFC1
Automatic basis set eextrapolation			
Extrapolate (n/m, bas)			Extrapolation of the basis set family “bas” (bas=cc,aug-cc, cc-core, ano, saug-ano, aug-ano, def2; if omitted “cc-pVnZ” is used) for cardinal numbers n,m (n<m=2,3,4,5), e.g. Extrapolate(2/3,cc) extrapolates the SCF, MP2 and MDCI energies to the basis set limit. “core” refers to basis sets with core correlation function. In this case the frozen core approximation is - by default - turned off. This setting can be overridden in the “methods” block if one just wants to use the basis set with core correlation functions (steep primitives) but without unfreezing the core electrons.
Extrapolate (n, bas)			Calculate the first n-energies for member of the basis set family basis, e.g. Extrapolate(3) is doing calculations with cc-pVDZ, cc-pVTZ and cc-pVQZ.
ExtrapolateEP2 (n/m, bas,[method,method- details])			Similar: performs SCF, MP2 and MDCI calculations. The higher basis set can only be done with DLPNO-CCSD(T) or MP2 methods and then used to extrapolate the MDCI calculation to the basis set limit.
ExtrapolateEP3 (bas,[method,method- details])			Similar to EP2: for the high basis set method we go one cardinal number higher.
CASSCF related options. All of them require the CASSCF block as minimal input			
DMRG NEVPT2			Sets DMRG as “CISStep” in CASSCF SC NEVPT2

SC-NEVPT2 RI-NEVPT2 FIC-NEVPT2 DLPNO-NEVPT2 CASPT2 RI-CASPT2 DCD-CAS(2) RI-DCD-CAS(2)			SC-NEVPT2 same as NEVPT2 SC-NEVPT2 with the RI approximation FIC-NEVPT2 aka PC-NEVPT2 FIC-NEVPT2 in the framework of DLPNO FIC-CASPT2 FIC-CASPT2 with the RI approximation 2nd order Dynamic Correlation Dressed CAS 2nd order Dynamic Correlation Dressed CAS with RI approximation
(internally contracted)Multireference methods beyond NEVPT2/CASPT2: If specified in a single keyword all information about reference spaces, number of roots etc. is taken from the CASSCF module that is assumed to be run in advance. These methods reside in the autoci module. More refined settings require the autoci block in the input.			
FIC-MRCI FIC-DDCI3 FIC-CEPA0 FIC-ACPF FIC-AQCC		CIType	Invokes the fully internally contracted MRCI Fully internally contracted DDCI3 Fully internally contracted CEPA0 Fully internally contracted ACPF Fully internally contracted AQCC
(uncontracted)Multireference methods: If specified in a single keyword all information about reference spaces, number of roots etc. is taken from the CASSCF module that is assumed to be run in advance. In general, these calculations are of the individually selecting type and are very time consuming. Very many flags can be set and modified for these methods and in general using these methods requires expert users! In general see the variables Tsel, Tpre and That that define the individual selection process. All of these methods can be used with RI integrals by using RI-MRCI etc. However, then the calculations become even more time consuming since integrals are made one- by one on the fly. Non-RI calculations will be pretty much limited to about 200-300 orbitals that are included in the CI			
MRCI MRCI+Q MRACPF MRAQCC MRDDCI1 MRDDCI2 MRDDCI3 MRDDCI n +Q SORCI	MRCI	CIType	Initiates a multireference configuration interaction calculation with single and double excitations Same with multireference Davidson correction for unlinked quadruples Average coupled-pair functional Average quadratic coupled-cluster Difference dedicated CI with one degree of freedom Same with two degrees of freedom Same with three degrees of freedom MRDDCI with Davidson correction Spectroscopy oriented CI

Frozen core features. NOTE: this deviates from previous versions of ORCA! We are now counting core electrons rather than using an energy window. If you do want to use an orbital energy window use <code>%method FrozenCore FC_EWIN end</code> . Otherwise the EWin commands will be ignored! (alternatives are FC_ELECTRONS (default) and FC_NONE).			
FROZENCORE	METHOD	FrozenCore	Use a frozen core. By default this is done by counting the number of chemical core electrons
NOFROZENCORE			Do not use a frozen core
Semiempirical methods			
ZINDO/S			Selects the ZINDO/S method
ZINDO/1			Selects the ZINDO/1 method
ZINDO/2			Selects the ZINDO/2 method
NDDO/1			Selects the NDDO/1 method
NDDO/2			Selects the NDDO/2 method
MNDO			Selects the MNDO method
AM1			Selects the AM1 method
PM3			Selects the PM3 method
Algorithmic variations, options, add-ons, modifiers,...			
RHF or RKS	SCF	HFTYP	Selects closed-shell SCF
UHF or UKS			Selects spin unrestricted SCF
ROHF or ROKS			Selects open-shell spin restricted SCF
AllowRHF	METHOD	ALLOWRHF	Allow a RHF calculation even if the system is open-shell (Mult>1). Default is to switch to UHF then
RI	METHOD	RI	Sets RI=true to use the RI approximation in DFT calculations. Default to Split-RI-J
NORI			Sets RI=false
RIJCOSX	METHOD/ SCF	RI, KMatrix	Sets the flag for the efficient RIJCOSX algorithm (treat the Coulomb term via RI and the Exchange term via seminumerical integration)
RI-JK	METHOD/ SCF	RI, KMatrix	Sets the flag for the efficient RI algorithm for Coulomb and Exchange. Works for SCF (HF/DFT) energies and gradients. Works direct or conventional.
SPLITJ	SCF	JMATRIX	Select the efficient Split-J procedure for the calculation of the Coulomb matrix in non-hybrid DFT (rarely used)
SPLIT-RI-J	SCF	JMATRIX,RI	Select the efficient Split-RI-J procedure for the improved evaluation of the RI-approximation to the Coulomb-matrix
NoSplit-RI-J	SCF	JMATRIX,RI	Turns the Split-RI-J feature off (but does not set the RI flag to false!)

RI-J-XC	SCF	JMATRIX, KMA- TRIX,RI	Turn on RI for the Coulomb term <i>and</i> the XC terms. This saves time when the XC integration is significant but introduces another basis set incompleteness error. (rarely used)
DIRECT SEMIDIRECT CONV	SCF	SCFMODE	Selects an integral direct calculation Selects an integral semidirect calculation (rarely used nowadays) Selects an integral conventional calculation
NOITER	SCF	MAXITER	Sets the number of SCF iterations to 0. This works together with MOREAD and means that the program will work with the provided starting orbitals.
Initial guess options: In most cases the default PMODEL guess will be adequate. In some special situations you may want to switch to a different choice			
PATOM PMODEL HUECKEL HCORE MOREAD AUTOSTART NOAUTOSTART	SCF	GUESS AUTOSTART	Selects the polarized atoms guess Selects the model potential guess Selects the extended Hückel guess Selects the one-electron matrix guess Read MOs from a previous calculation (use %moinp "myorbitals.gbw" in a separate line to specify the GBW file that contains these MOs to be read) Try to start from the existing GBW file of the same name as the present one Don't try to do that
Basis-set related keywords			
DecontractBas NoDecontractBas DecontractAuxJ NoDecontractAuxJ DecontractAuxJK NoDecontractAuxJK DecontractAuxC NoDecontractAuxC Decontract	BASIS	DecontractBas NoDecontractBas DecontractAuxJ NoDecontractAuxJ DecontractAuxJK NoDecontractAuxJK DecontractAuxC NoDecontractAuxC Decontract	Decontract the basis set. If the basis set arises from general contraction, duplicate primitives will be removed. Do not decontract the basis set Decontract the AuxJ basis set Do not decontract the AuxJ basis Decontract the AuxJK basis set Do not decontract the AuxJK basis Decontract the AuxC basis set Do not decontract the AuxC basis Decontract all (orbital and auxiliary) basis sets
Relativistic options: There are several variants of scalar relativistic Hamiltonians to use in all electron calculations			
DKH or DKH2	REL	METHOD/ORDER	Selects the scalar relativistic Douglas-Kroll-Hess Hamiltonian of 2nd order

ZORA	REL	METHOD	Selects the scalar relativistic ZORA Hamiltonian
ZORA/RI	REL	METHOD	Selects the scalar relativistic ZORA Hamiltonian in RI approximation
IORA/RI	REL	METHOD	Selects the scalar relativistic IORA Hamiltonian in RI approximation
IORAmm/RI	REL	METHOD	Selects the scalar relativistic IORA mm (modified metric) Hamiltonian in RI approximation
Grid options			
GRID n ($n = 0-7$) FINALGRID n	METHOD	GRID	Selects the DFT integration grid no n Selects the DFT final integration grid no n
NOFINALGRID			Turns the final grid feature off
GRIDX n ($n = 1-9$) NOFINALGRIDX	METHOD	GRIDX	Grids for the COSX approximation. A sequence of three grids is used. Higher accuracy at higher cost is offered by the higher grids. Turn off the final grid in COSX (not recommended)
Convergence thresholds: These keywords control how tightly the SCF and geometry optimizations will be converged. The program makes an effort to set the convergence thresholds for correlation modules consistently with that of the SCF.			
NORMALSCF LOOSESCF SLOPPYSCF STRONGSCF TIGHTSCF VERYTIGHTSCF EXTREMESCF SCFCONV n	SCF	CONVERGENCE	Selects normal SCF convergence Selects loose SCF convergence Selects sloppy SCF convergence Selects strong SCF convergence Selects tight SCF convergence Selects very tight SCF convergence Selects “extreme” convergence. All thresholds are practically reduced to numerical precision of the computer. Only for benchmarking (very expensive). Selects energy convergence check and sets $ETol$ to 10^{-n} ($n = 6-10$). Also selects appropriate <i>thresh</i> , <i>tcut</i> , and <i>bfcut</i> values.
VERYTIGHTOPT TIGHTOPT NORMALOPT LOOSEOPT			GEOM

Convergence acceleration: the default is DIIS which is robust. For most closed-shell organic molecules SOSCF converges somewhat better and might be a good idea to use. For “trailing convergence”, KDIIS or the true second-order procedures NRSCF and AHSCF might be good choices.			
DIIS	SCF	DIIS	Turns DIIS on
NODIIS			Turns DIIS off
KDIIS	SCF	KDIIS	Turns Kollmar’s DIIS on
NRSCF	SCF	NR	Turns Newton-Raphson SCF on
AHSCF			Same but with augmented Hessian step
NONRSCF			Turns Newton-Raphson SCF off
SOSCF	SCF	SOSCF	Turns SOSCF on
NOSOSCF			Turns SOSCF off
DAMP	SCF	CNVDAMP	Turns damping on
NODAMP			Turns damping off
LSHIFT	SCF	CNVSHIFT	Turns level shifting on
NOLSHIFT			Turns level shifting off
Convergence strategies (does not modify the convergence criteria)			
EasyConv			Assumes no convergence problems.
NormalConv			Normal convergence criteria.
SlowConv			Selects appropriate SCF converger criteria for difficult cases. Most transition metal complexes fall into this category.
VerySlowConv			Selects appropriate SCF converger criteria for very difficult cases.
ForceConv			Force convergence: do not continue with the calculation, if the SCF did not fully converge.
IgnoreConv			Ignore convergence: continue with the calculation, even if the SCF wavefunction is far from convergence.
CPCM(solvent)	CPCM		Invoke the conductor-like polarizable continuum model with a standard solvent (see section 9.35 for a list of solvents). If no solvent is given, infinity (a conductor) is assumed.
C-PCM			
Spin-orbit coupling			
SOMF(1X)	REL	SOCType, SOCFlags	Invokes the RI-SOMF(1X) treatment of the spin-orbit coupling operator
Miscellaneous options			
ANGS	COORDS	UNITS	Select angstrom units
BOHRS			Select input coordinates in atomic units
FRACOCC	SCF	FRACOCC	Turns the fractional occupation option on (FOD is always calculated in this case)

SMEAR	SCF	SMEARTEMP	Temperature for occupation number smearing on (default is 5000 K; FOD (see 9.6.8.2) is always calculated in this case)
NOSMEAR			Turn occupation number smearing off
KEEPINTS	SCF	KEEPINTS	Keep two electron integrals on disk
NOKEEPINTS			Do not keep two electron integrals
KEEPDENS	SCF	KEEPDENS	Keep the density matrix on disk
NOKEEPDENS			Do not keep the density matrix
READINTS	SCF	READINTS	Reading of two electron integrals on
NOREADINTS			Reading of two electron integrals off
CHEAPINTS	SCF	USECHEAPINTS	Use the cheap integral feature in direct SCF calculations
NOCHEAPINTS			Turn that feature off
FLOAT	SCF	VALFORMAT	Set storage format for numbers to single precision (SCF, RI-MP2, CIS, CIS(D), MDCI)
DOUBLE	SCF	VALFORMAT	Set storage format for numbers to double precision (default)
UCFLOAT	SCF	VALFORMAT	Use float storage in the matrix containers without data compression
CFLOAT	SCF	VALFORMAT	Use float storage in the matrix containers with data compression
UCDOUBLE	SCF	VALFORMAT	Use double storage in the matrix containers without data compression
CDOUBLE	SCF	VALFORMAT	Use double storage in the matrix containers with data compression
Output control			
NORMALPRINT	OUTPUT	PRINTLEVEL	Selects the normal output
MINIPRINT			Selects the minimal output
SMALLPRINT			Selects the small output
LARGEPRINT			Selects the large output
PRINTMOS	OUTPUT	Print[p_MOS]	Prints MO coefficients
NOPRINTMOS	OUTPUT		Suppress printing of MO coefficients
PRINTBASIS	OUTPUT	Print[p_basis]	Print the basis set in input format
PRINTGAP	OUTPUT	Print[p_homolumogap]	Prints the HOMO/LUMO gap in each SCF iteration. This may help to detect convergence problems
ALLPOP	OUTPUT	Print[...]	Turns on all population analysis
NOPOP			Turns off all population analysis
MULLIKEN			Turns on the Mulliken analysis
NOMULLIKEN			Turns off the Mulliken analysis
LOEWDIN			Turns on the Loewdin analysis
NOLOEWDIN			Turns off the Loewdin analysis
MAYER			Turns on the Mayer analysis

NOMAYER NPA NBO NONPA NONBO REDUCEDPOP NOREducedPOP			Turns off the Mayer analysis Turns on interface for the NPA analysis using the GENNBO program Turns on the interface for the NPA plus NBO analysis with the GENNBO program Turns off NPA analysis Turns off NBO analysis Prints Loewdin reduced orb.pop per MO Turns this feature off
UNO	SCF	UNO	Produce UHF natural orbitals
AIM			Produce a WFN file
XYZFILE PDBFILE	OUTPUT	XYZFILE PDBFILE	Produce an XYZ coordinate file Produce a PDB file

Compression and storage. The data compression and storage options deserve some comment: in a number of modules including RI-MP2, MDCI, CIS, (D) correction to CIS, etc. the program uses so called “Matrix Containers”. This means that the data to be processed is stored in terms of matrices in files and is accessed by a double label. A typical example is the exchange operator $\mathbf{K}^{ij}$ with matrix elements $K^{ij}(a,b) = (ia|jb)$. Here the indices i and j refer to occupied orbitals of the reference state and a and b are empty orbitals of the reference state. Data of this kind may become quite large (formally N^4 scaling). To store the numbers in single precision cuts down the memory requirements by a factor of two with (usually very) slight loss in precision. For larger systems one may also gain advantages by also compressing the data (e.g. use a “packed” storage format on disk). This option leads to additional packing/unpacking work and adds some overhead. For small molecules UCDOUBLE is probably the best option, while for larger molecules UCFLOAT or particularly CFLOAT may be the best choice. Compression does not necessarily slow the calculation down for larger systems since the total I/O load may be substantially reduced and thus (since CPU is much faster than disk) the work of packing and unpacking takes less time than to read much larger files (the packing may reduce disk requirements for larger systems by approximately a factor of 4 but it has not been extensively tested so far). There are many factors contributing to the overall wall clock time in such cases including the total system load. It may thus require some experimentation to find out with which set of options the program runs fastest with.

! CAUTION !

- It is possible that FLOAT may lead to unacceptable errors. Thus it is not the recommended option when MP2 or RI-MP2 gradients or relaxed densities are computed. For this reason the default is DOUBLE.
- If you have convinced yourself that FLOAT is OK, it may save you a factor of two in both storage and CPU.

Global memory use. Some ORCA modules (in particular those that perform some kind of wavefunction based correlation calculations) require large scratch arrays. Each module has an independent variable